Planning the Flight

Week 6 — Tuesday Lecture

Today

- Why a bad flight cannot be fixed by software
- How high, how much overlap, what pattern, which way to point the camera
- Getting the **sides** of a building into the frame
- Hard constraints first, and flight time by hand
- The class flight planner you will use Thursday

The photos are the ceiling

- Every product — orthomosaic, surface, 3D model — is built from the photos you bring back
- Processing can only find what the camera saw, as sharply as it saw it
- Planning decides the pixel size, the overlap, the pattern, and how long you are in the air

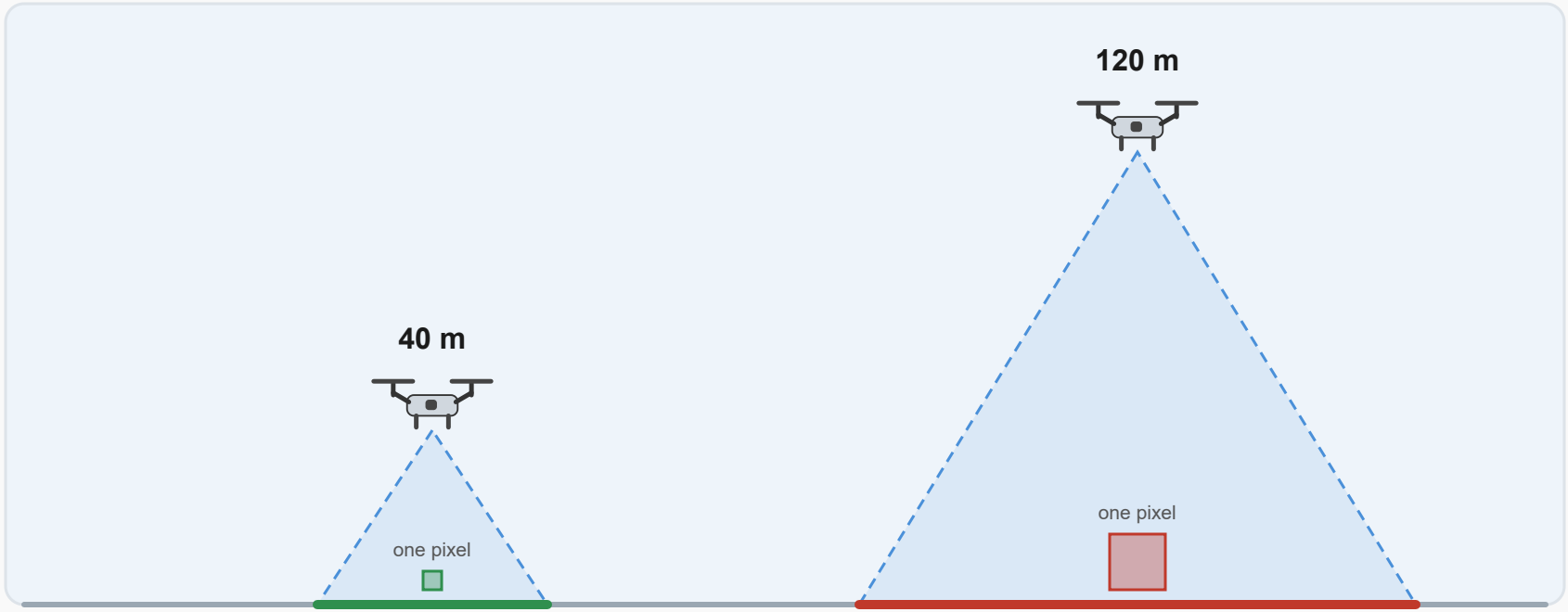
Thursday you plan four missions in the class planner. Today is why the numbers are what they are.

What is the smallest thing this map has to show?

Answer that first. Every other number follows from it.

Altitude decides detail, and how long you fly

Double the height and every pixel covers twice as much ground, but each photo covers four times the area.



A 5 mm crack needs roughly 1 cm pixels to show up at all. No amount of processing recovers detail you never captured.

Flying twice as high doubles the size of every pixel, but each photo covers four times the area.

Height is the dial

- **Ground sample distance (GSD):** the real-world size of one pixel
- A 3-in divot at 1.5 cm per pixel is about five pixels across — visible. At 5 cm it is one pixel — gone.
- Choose the GSD that resolves the smallest feature, then fly at the **highest** altitude that still delivers it

Lower is not better. Lower is more photos, more lines, more batteries, and more hours of processing.

The tradeoff game

A 50-acre site. You need drainage **and** 5 mm cracks.

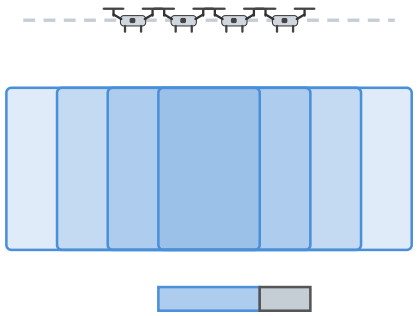
20 minutes at 120 m (4 cm GSD), or 60 minutes at 40 m (1 cm GSD). Your battery lasts 25 minutes. What is your plan?

Why overlap is the number that matters

Overlap exists so the same point lands in several photos. Without that, there is nothing to triangulate.

Forward overlap, 80%

between one photo and the next



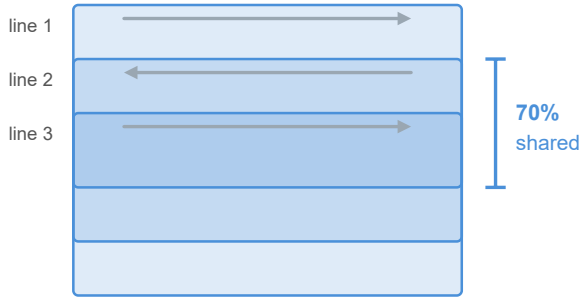
80% of each photo repeats the one before

only the grey 20% is new ground

so every point on the ground is caught by
5 to 9 different photos.

Side overlap, 70%

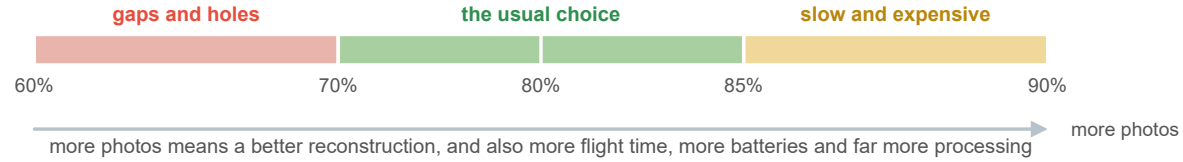
between one flight line and the next



70% of each strip is photographed twice.

Too little sidelap leaves gaps between lines
that no software can fill in afterwards.

What you are trading



more photos means a better reconstruction, and also more flight time, more batteries and far more processing

Overlap exists so the same point lands in several photos. Without that, there is nothing to triangulate.

Forward and side overlap do not cost the same

Both give you more images to store and process. Only one of them makes you fly further.

More forward overlap



The same flight lines, flown the same way.
The camera simply fires more often.

Flight time: barely changes
Batteries: the same
You pay in: storage and processing

More side overlap



The lines sit closer together, so the site needs more of them to be covered.

Flight time: up with every extra line
Batteries: more of them
You pay in: time, batteries, storage, processing

So push forward overlap when in doubt; it is the cheap one. Adding side overlap costs you flight lines, and flight lines cost batteries. Overlap is a design decision, not a setting to max out.

More forward overlap is more photos on the same lines. More side overlap is more lines, and every line costs battery.

How much overlap

Site	Forward	Side
Open, flat, textured ground	70 %	60 %
General mapping	**80 %**	**70 %**
Tall vegetation, complex structures	85 %	80 %
Repeating patterns: parking lots, row crops, grass	85 %	80 %

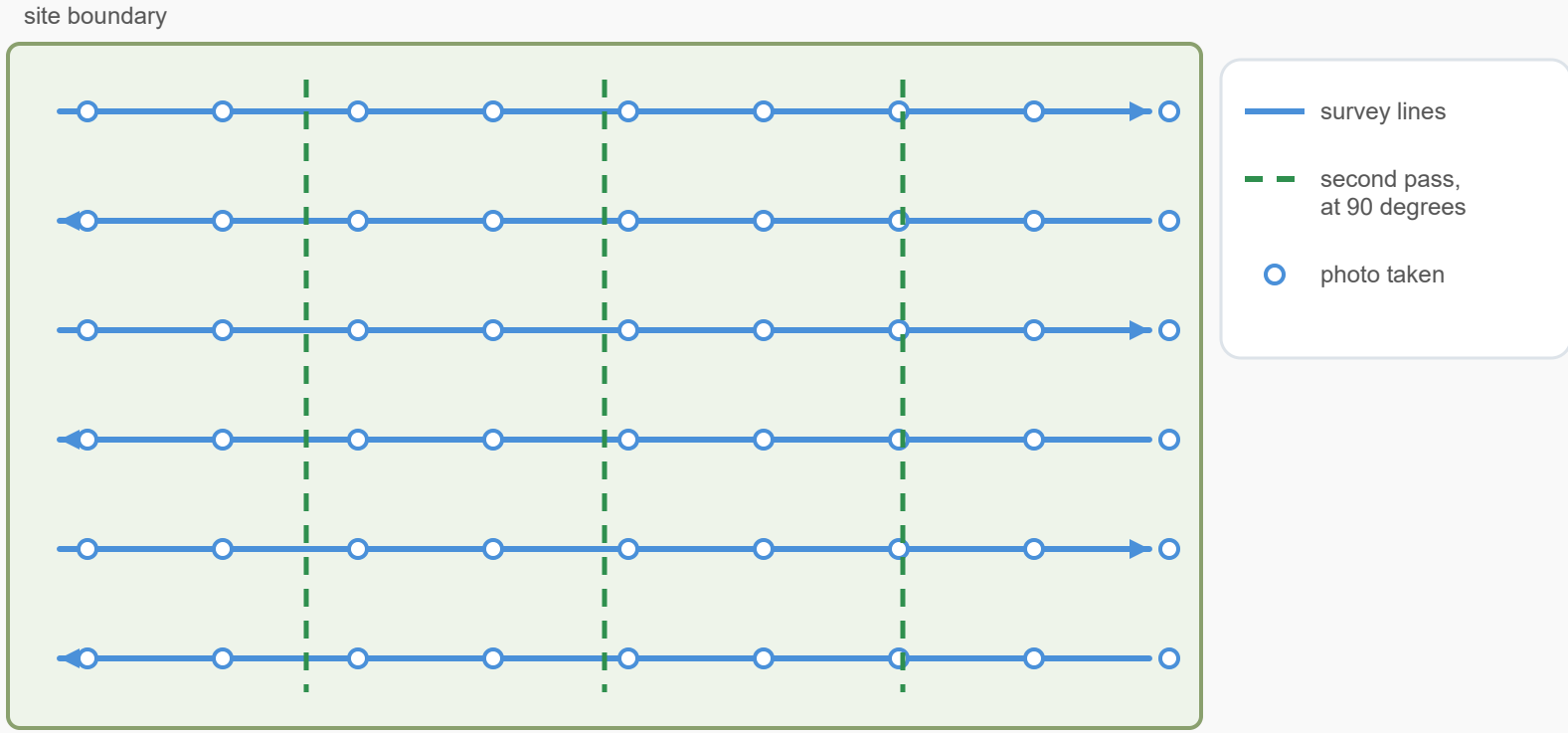
Above about 85 % there is little left to gain.

Which overlap is the cheap one?

Forward or side? Why does it matter which one you raise first?

A mapping grid, and when to add a second pass

Lines run one way for the survey. A perpendicular pass over the same ground fixes what a single direction cannot.



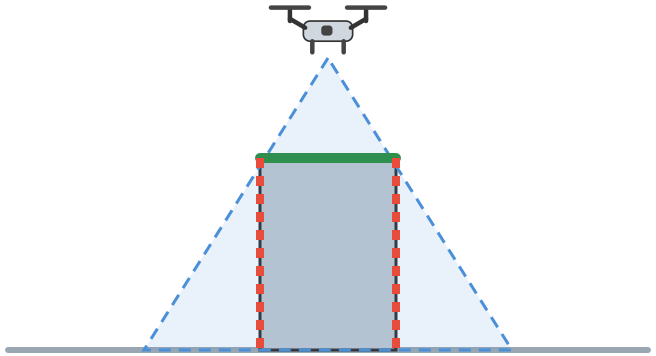
Flat, open ground rarely needs the second pass. Tall trees, buildings and steep slopes usually do.

Parallel lines, alternating direction. A second pass at ninety degrees only when the site needs it.

Nadir, oblique, and why you often fly both

A camera pointing straight down never sees a wall. That is the whole reason oblique passes exist.

Nadir, camera at 0

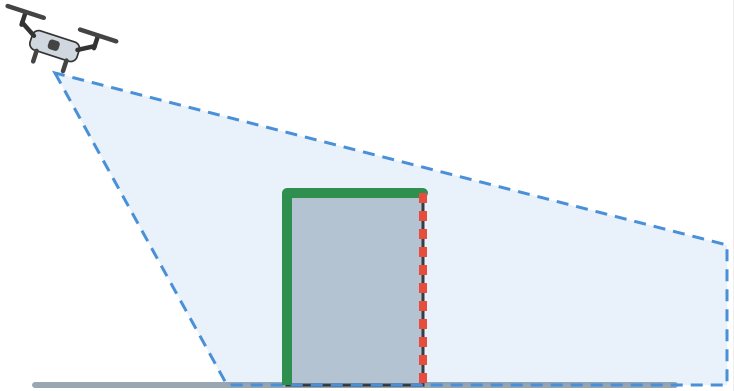


roof captured, walls missed

green: surfaces the camera sees

red dashed: surfaces it cannot

Oblique, camera at 30 to 45



roof and the near wall captured

green: surfaces the camera sees

red dashed: surfaces it cannot

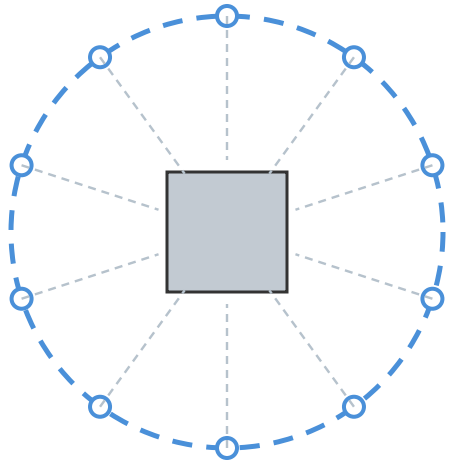
One oblique pass still leaves three walls unseen. Orbit the structure, or fly obliques from four directions.

A nadir camera sees the roof and nothing else. Tilting it to 30 or 45 degrees brings the near wall into view.

Orbiting a structure

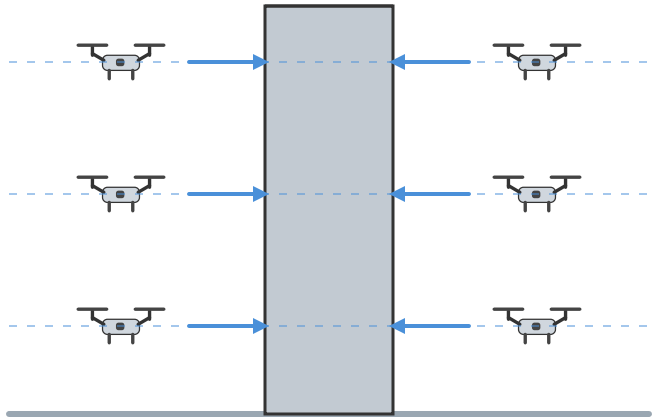
A grid maps the ground. Vertical faces need the camera to come round the side, usually at more than one height.

From above



camera always pointed inward

From the side



one ring per height, so no band of wall is missed

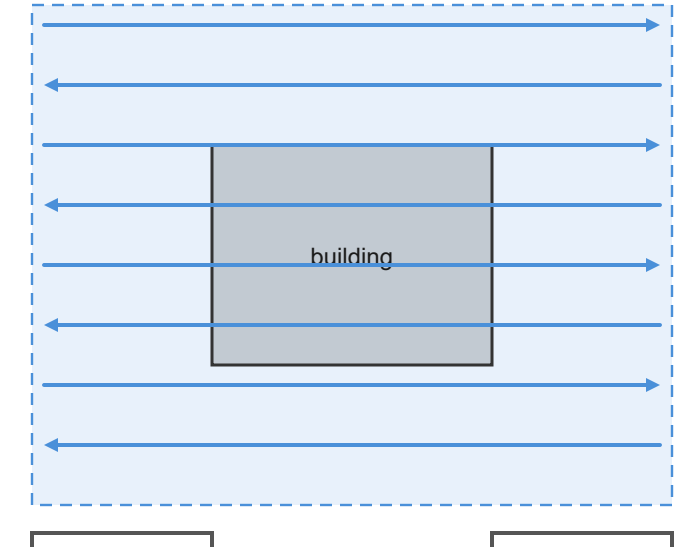
A single orbit at one height leaves the top and bottom of a tall structure poorly reconstructed.

The camera points inward and circles the structure. Tall structures need a ring at more than one height.

Plan for the sides from the start

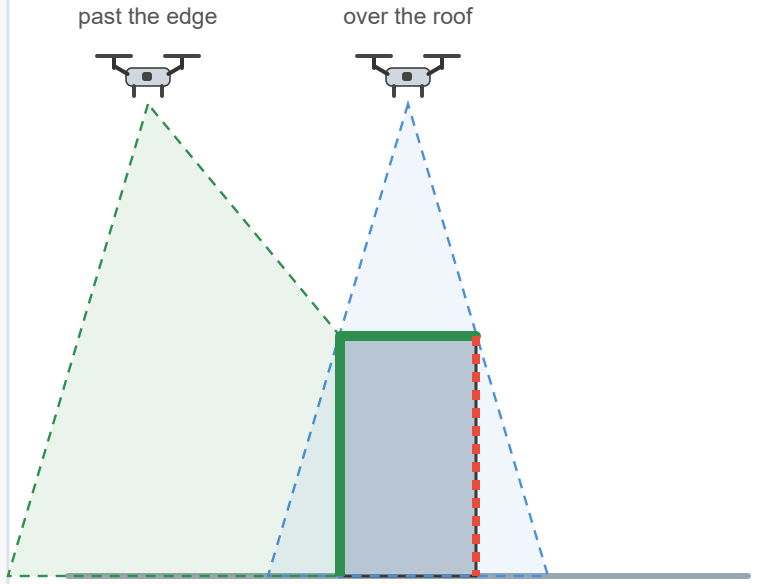
A grid that stops at the footprint sees only the roof. Run the lines well past the building, or add a line the planner will not draw for you.

From above



lines run a building height or more past every edge

From the side



the edge of the frame catches the wall, even at nadir

Water towers, bridges, and anything you want as a 3D object: same rule, every side. Or add an oblique line or an orbit outside the planner.

A grid that stops at the footprint sees only the roof. Lines a building height past every edge catch the walls at the edge of the frame.

Plan for the sides from the start

If the goal is a **3D object** — a building, a water tower, a bridge pier — decide how you get its sides *before* you draw the area:

1. **Run the lines well past it.** A line or two beyond the building on every side, at least one building height out. Cheap, and the planner will do it if the area is drawn large enough.
2. **Add a dedicated pass the planner will not draw.** An oblique line along each face, or an orbit, flown after the grid.

A map of the roof is not a model of the building.

Hard constraints first

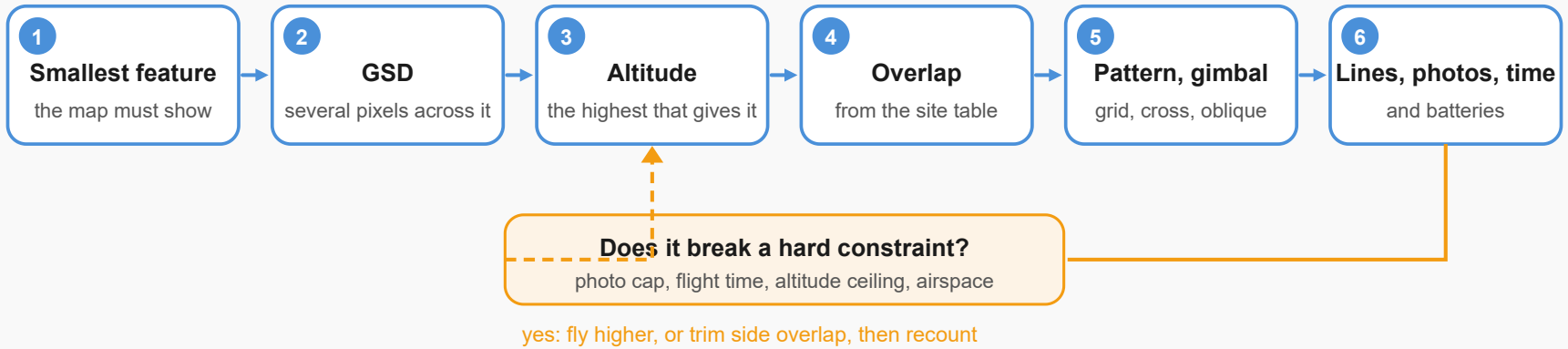
Thursday's missions each have one line you cannot cross:

- **99 photos** — the DJI Fly limit
- **12 minutes** of mapping time
- **400 ft AGL**, and the **airspace ceiling** near an airport
- A **GSD** the smallest feature demands

Everything else is a goal. Start from the constraint that bites, and plan outward from it.

Start from what the map must show, not from the drone

Each choice follows from the one before it. When the plan breaks a hard constraint, go back one step, not to the start.



The planner draws the lines and counts the photos. The six choices are still yours.

Start from what the map must show and work right. When a hard constraint breaks, go back one step, not to the start.

Flight time by hand

The planner reports path distance. Minutes and batteries are your arithmetic, and the hard constraints are written in minutes.

Minutes from distance and speed

$$\text{time (min)} = \frac{\text{path distance (ft)}}{\text{speed (mph)} \times 88}$$

88 ft per minute for every 1 mph

(metric: metres ÷ (m/s × 60))

5,280 ft at 10 mph

5,280 ÷ (10 × 88) = 6.0 min

plus turns, climb, and the trip to the first line

Batteries from minutes

$$\text{batteries} = \frac{\text{flight time (min)}}{\text{battery (min)} - \text{reserve}}$$

round up; a part battery is a whole battery

keep a landing reserve of about 20 %

34 min of lines, 25-min battery

34 ÷ (25 − 5) = 1.7 → 2 batteries

and a battery swap means a pause in the pattern

A 12-minute limit is a distance limit in disguise: at 10 mph it is 10,560 ft of flight line, and not one foot more.

The planner gives you distance. Minutes and batteries are your arithmetic.

Flight time by hand

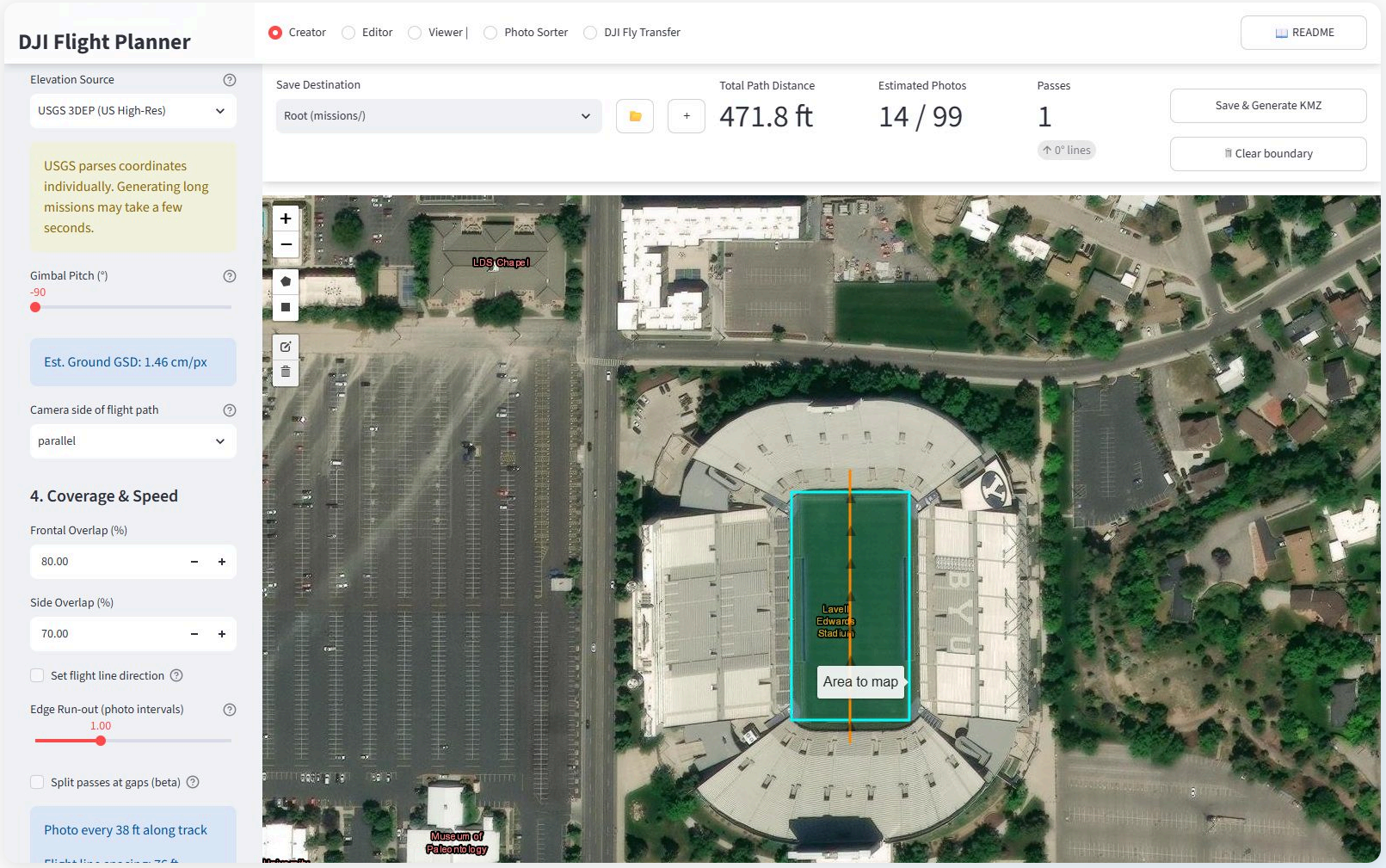
$$\text{time (min)} = \frac{\text{path distance (ft)}}{\text{speed (mph)} \times 88}$$

5,280 ft of flight line at 10 mph:

$$\frac{5{,}280}{10 \times 88} = 6.0 \text{ min}$$

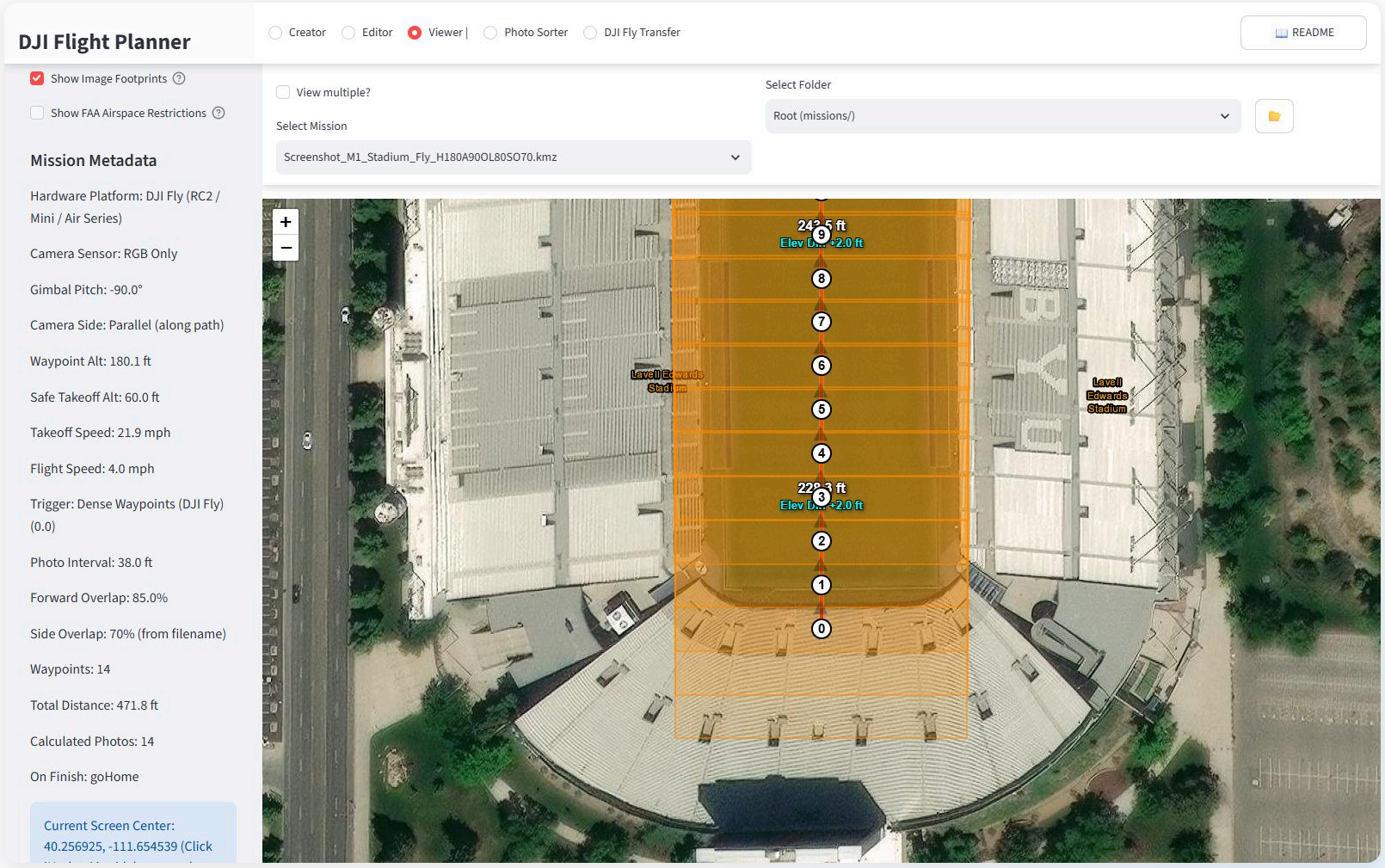
Add a minute or two for turns and the climb. A 12-minute limit at 10 mph is **10,560 ft of line, and not one foot more.**

The class flight planner



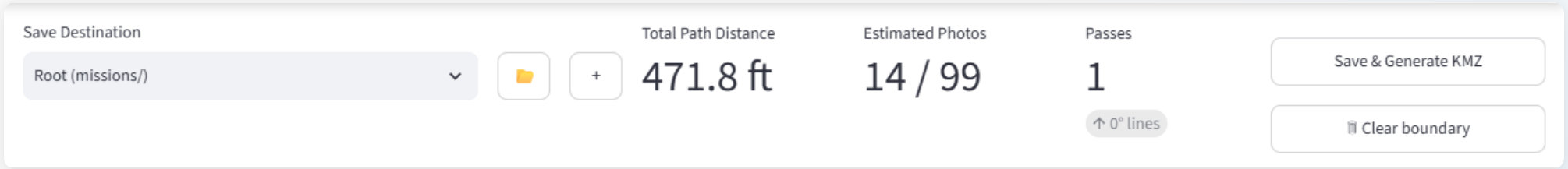
Creator tab: switch to a mapping mission, pick the drone platform, draw the area, set altitude, overlap, speed and gimbal.

The lines it draws



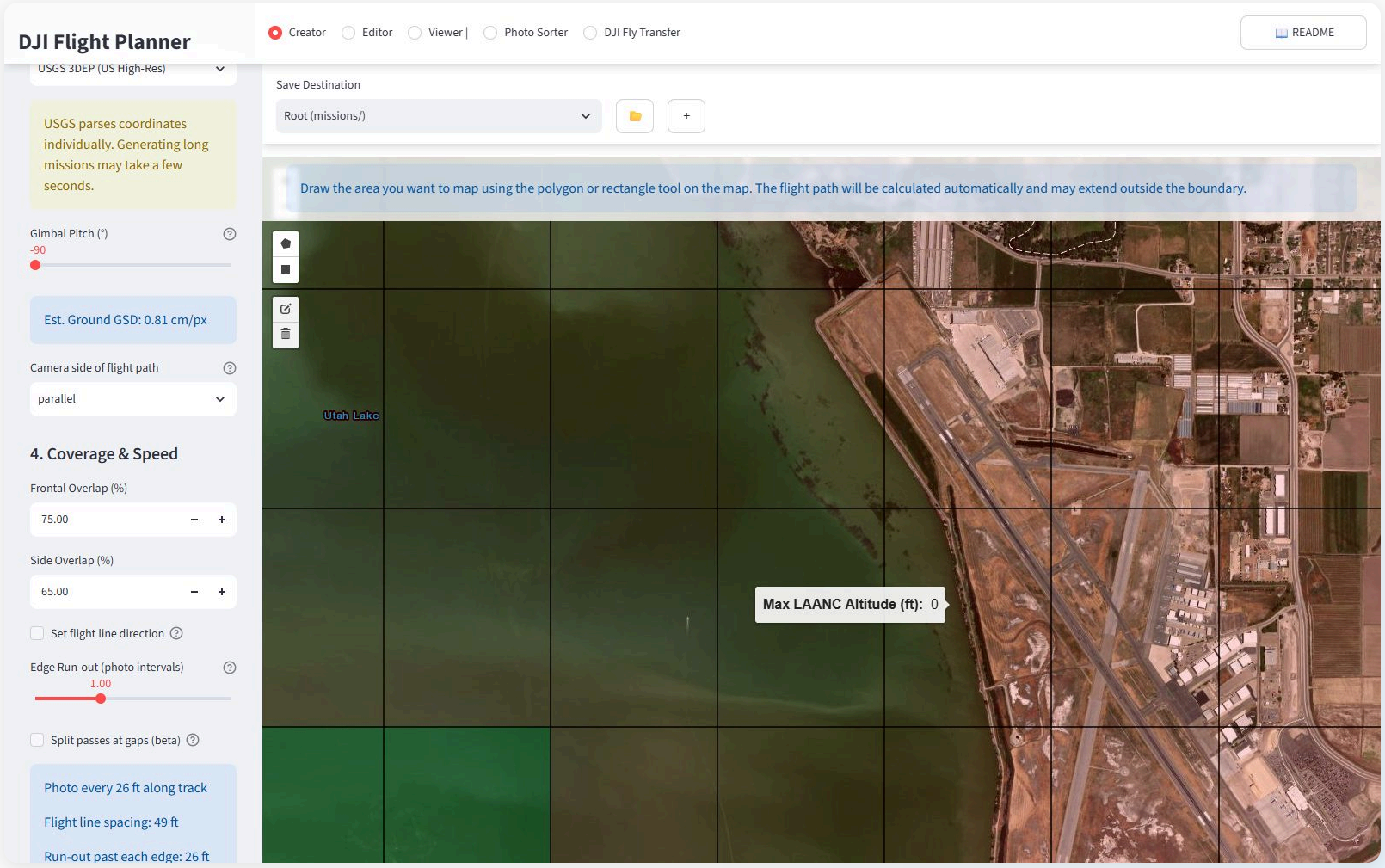
Viewer tab: the grid over the site. Check that the lines run past anything you need the sides of.

What it reports — and what it does not



Path distance, photo count against the cap, and passes. GSD is in the sidebar. Flight time is yours to compute from the distance.

The airspace layer

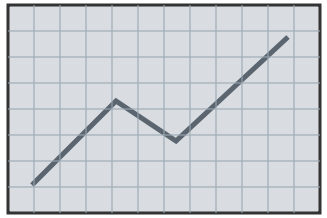


Beside the Provo airport the ceiling reads **0 ft**: no flight there at all. Read the cell you will actually fly in, not the one next to it.

Resolution and accuracy are different questions

A map can show tiny detail and still sit in the wrong place. Only a surveyed point tells you which you have.

Resolution

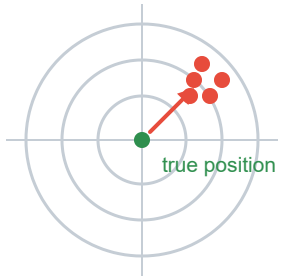


Can you see it at all?

Set by GSD. A 5 mm crack needs pixels of about 1 cm to appear.

Fly lower to improve it.

Accuracy



Is it in the right place?

Tight measurements can still all be shifted together, and look convincing.

Flying lower does not fix this.

Ground truth



surveyed targets on the ground

How would you know?

Ground control points fix the map to real coordinates. Extra checkpoints, held back, prove how close you got.

Without ground control, a map with 1 cm pixels can sit several metres from where it belongs.

Three different questions. Flying lower answers the first one only.

How you will know it worked

- **Resolution** comes from GSD and answers "can I see it at all"
- **Accuracy** is how close the map sits to the truth. Flying lower does not fix it
- **Ground control points** tie the map to real coordinates. **Checkpoints**, held back, prove how close you got

A 1 cm map can sit several metres from where it belongs and look perfect doing it. Week 7 is where you place the targets.

What to do Thursday

Mission	The constraint that bites
Stadium turf check	99 photos, GSD ≤ 1.5 cm
Inspecting the Y	Steep slope, oblique gimbal, GSD ≤ 2 cm
Racing the sunset	12 minutes, and the airport ceiling
Rock Canyon Park (team)	Your drone, your controller, not flat

Fill in the Mission Log as you go. Lab page: Creating Flight Plans.